

Projection Mapping in the Light: A Preliminary Attempt to Substitute Projectors for Room Lights

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ABSTRACT

Projection mapping (PM) in a bright room creates the problem of reducing the contrast of the projected texture because ambient lighting elevates the black level of the projection target. In this paper, we developed a pseudo-ambient lighting technique that turns off the original ambient lighting and reproduces its illumination from the projectors on surfaces other than the target. We confirmed that the proposed technique could reproduce a bright room while suppressing the contrast reduction of the projected texture on the target, which helped to improve the viewing experience.

Index Terms: Human-centered computing—Human computer interaction (HCI)—Interaction paradigms—Mixed / augmented reality; Computing methodologies—Computer graphics—Graphics systems and interfaces—Perception

1 INTRODUCTION

Projection mapping (PM) is a technology that allows computer-generated images to be superimposed on a target object in real space using a projector. A crucial assumption of PM is that images are projected in dark rooms to suppress the contrast reduction. When bright ambient lighting illuminates the projected object, its black level is elevated, so the contrast of the projected texture is significantly reduced. As a result, the naturalness of the projected texture may be spoiled, reducing viewing experience. However, ambient lighting cannot be subtracted, so a target appearance that is darker than the surface appearance under the original ambient lighting cannot be reproduced.

This paper proposes a technique to suppress the contrast reduction of projected content by substituting projectors for the ambient lighting. In contrast to conventional PM, which projects images only on the target object to change its texture, our PM technique uses projectors also as ambient light sources. Thereby, we propose a pseudo-ambient lighting technique that turns off the original ambient lighting and reproduces its illumination effect on surfaces other than the projection target. This enables PM with minimal contrast reduction of the projected texture while illuminating the scene as if the original ambient lighting is turned on.

Researchers have found PM to be useful in various application fields, such as product and architecture design [2]. However, the

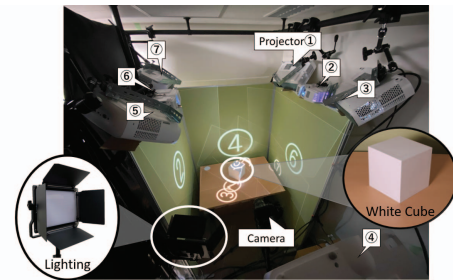


Figure 1: Installation of projectors and other equipment

appearance of the projected object is still not visually satisfactory in terms of naturalness, because surfaces of physical objects other than the projected object are in the dark. Our proposed method has the potential to solve this problem because it provides realistic product design in an environment closer to actual conditions.

2 PROPOSED METHOD

First, a camera captures images of a room under the original ambient lighting and measures the color information of surfaces in the room. Then, the ambient lighting is turned off. Based on the measured color information, we calculate a projection image such that the appearance of the room under the original ambient lighting is reproduced once it is projected. Multiple projectors are used to cover all the surfaces. In this ambient lighting reproduction, the projection target object is not lit by the projectors, so it becomes a black missing spot in the room. We then project target textures onto the target object to modulate its appearance. As a result, we achieve

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(a) Bright room condition under ambient lighting

(b) Emulated bright room condition under pseudo-ambient lighting

Figure 2: Results of brick texture projection

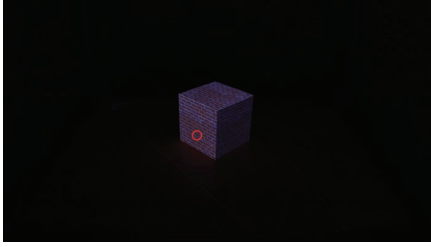


Figure 3: Result of brick texture projection in dark room condition

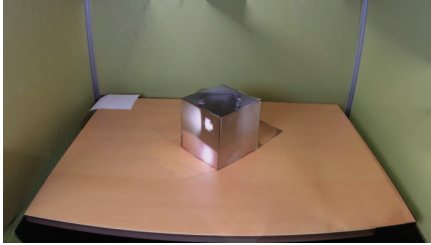


Figure 4: Results of specular reflection texture (environment mapping) projection

PM that changes the surface appearance of the target object while suppressing contrast reduction.

3 IMPLEMENTATION OF THE PROPOSED SYSTEM

Seven projectors (3×acer H6517ST, 4×BenQ TK850) were used to reproduce ambient lighting. We built a small room using green portable walls and a wood-grained desk, as shown in Fig. 1. A white cube was placed on the desk as the projection object. A camera with a fisheye lens (camera: canon EOS M6 Mark II, lens: canon EF8-15mm F4L Fisheye USM) was installed in a position where the entire small room could be photographed. Three of the projectors are used for outward-facing projection, mainly on the walls. The other four projectors are used for inward-facing projection, mainly on the desk and the projection object. We installed an LED array as ambient lighting in front of the room. The target appearance for the projection object was a brick texture.

To create the pseudo-ambient lighting, we applied geometric correction to align the projected images with the surfaces, and color compensation to reproduce the surface color under the original ambient lighting [3]. We also applied gamma correction and luminance attenuation of the overlapping projection area [1].

4 EXPERIMENT

4.1 Texture manipulation of the projection target

Fig. 2(a) and (b) show the projected results of the brick texture under the original ambient lighting and the pseudo-ambient lighting generated by the above process, respectively. In Fig. 2(a), the contrast of the texture is extremely reduced. On the other hand, in Fig. 2(b), the texture is clearly visible. The contrast reduction of the texture on the target is suppressed. Fig. 3 shows the projected result of the texture under dark room conditions. We measured the color information of the texture on the projection object in Fig. 2(a), 2(b), and Fig. 3. The red circle in Fig. 3 indicates the colorimetric area. Color difference in CIE 1976(ΔE) between Fig. 3 and Fig. 2(a) was 65.4. The difference between Fig. 3 and Fig. 2(b) was 27.5.

We also tried to switch the surface of the cube to a specular material. Specifically, we applied environment mapping to generate the projection image for the cube. The environment map was captured

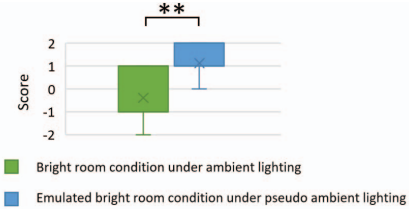


Figure 5: Results of the Naturalness Rating Questionnaire

using the fisheye camera. The projected result is shown in Fig. 4. The white cube has a diffuse reflection of lighting on its surface, but by projecting a texture that achieves the appearance of specular reflection, the object appeared to have a mirrored surface.

The projected result of an environment map under dark room condition does not look natural. If the surroundings are dark, the environment itself is invisible, so the projected environment map cannot be perceived as a mirror surface. On the other hand, with our proposed method, we project the environment map with pseudo-ambient lighting. It is suggested that the proposed method can reproduce texture information that cannot be expressed in a dark room.

4.2 Subjective evaluation of naturalness of appearance

We subjectively evaluated the naturalness of the PM results through a user study between PM of the object under the original ambient lighting and PM of the object under pseudo-ambient lighting by the proposed method. We recruited 15 participants (14 males and 1 female) between the ages of 21 and 25 years. The participants were asked to enter the darkened room of the experimental facility and were shown two different brick texture projections, one with the original ambient lighting and the other with the pseudo-ambient lighting (Fig. 2(a) and Fig. 2(b)). They were asked whether the brick texture looked natural in each condition. They answered on a 5-point scale from -2 to 2, with 2 indicating that the texture looked natural, 0 being undecided, and -2 indicating that the texture did not look natural. The results are shown in Fig. 5. The mean values were -0.4 for the ambient lighting condition and 1.1 for the pseudo-ambient lighting condition. A Wilcoxon signed-rank test showed that there was a significant difference ($p < .01$) between ambient lighting and pseudo-ambient lighting. Therefore, we confirmed that the brick textures appeared significantly more natural under pseudo-ambient lighting than under the original ambient lighting. This indicates that the proposed PM method helps to improve the naturalness of the projection mapping.

5 CONCLUSION

In this paper, we reproduced ambient lighting while using PM on an object. We confirmed that the contrast reduction of the texture on the target is suppressed by our method and that helps to improve the naturalness of PM. In the future, we will improve the color compensation and blending algorithms to realize the system for complex and large indoor environments.

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